

Chapter 7

Tools for Artificial Intelligence

In this chapter we will talk about the tools which are required to make project in artificial intelligence. All this bellow discussed tools are uses python programming language for making any project in any respective field. We have discussed about how anyone can install and use the tools to built a real time project in artificial Intelligence.

Using Python IDE

To download and install Python visit the official website of Python <https://www.python.org/downloads/> and choose your version. We have chosen Python version as per requirement all the codes in this book were on python version 3.6 as you can see in the Fig. 7.1.



Fig. 7.1 Python Download

Once the download is complete, run the exe for install Python. Now click on Install Now as in Fig. 7.2.



Fig. 7.2 Installation of Python

You can see Python installing at this point as in Fig. 7.3

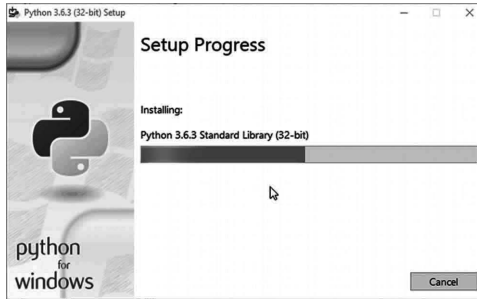


Fig. 7.3 Installation process

When it finishes, you can see a screen that says the Setup was successful. Now click on “Close” as in Fig. 7.4.



Fig. 7.4 Installation Complete

Using Anaconda

On windows:

Go to the following link: Anaconda.com/downloads.

The Anaconda Downloads Page will look something like this in Fig. 7.5:

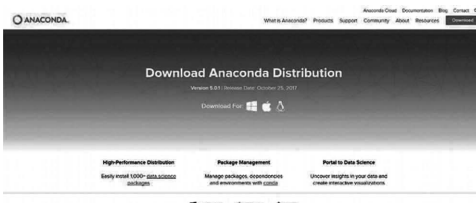


Fig. 7.5 Installation Complete

Select which ever your operating system is from the three listed operating systems as in Fig. 7.6. we have chosen windows.



Fig. 7.6 Anaconda Versions Select windows

Download the most recent Python 3 release as in Fig. 7.7. At the time of writing, the most recent release which I was using is Python 3.6 Version. Python 2.7 is legacy

Python. For problem solvers, select the Python 3.6 version. If you are unsure if your computer is running a 64-bit or 32-bit version of Windows, select 64-bit as 64-bit Windows is most common.

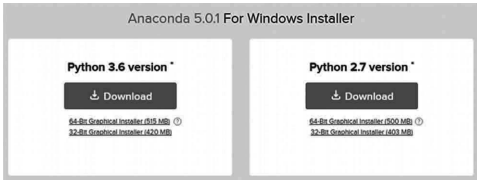


Fig. 7.7 Python version for Anaconda

The download is quite large so it may take a while for Anaconda to download. Once the download completes, open and run the .exe installer. At the beginning of the install, you need to click Next to confirm the installation as shown in Fig. 7.8.



Fig. 7.8 Python version for Anaconda

Then agree to the license as in Fig. 7.9.

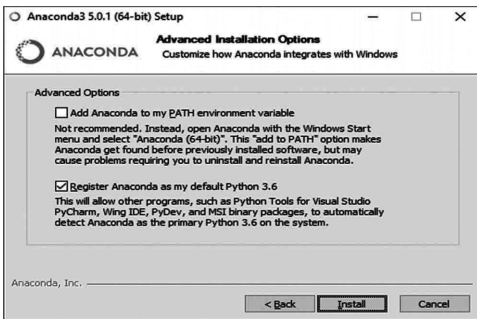


Fig. 7.9 License Agreement

At the Advanced Installation Options screen, I recommend that you do not check “Add Anaconda to my PATH environment variable” as shown in Fig. 7.10

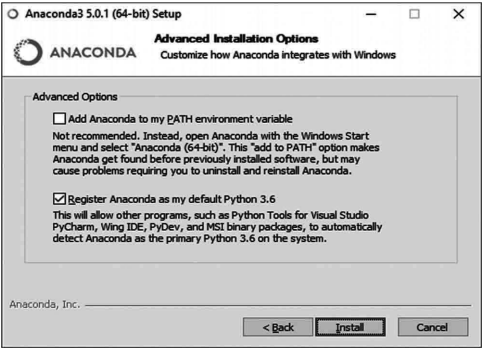


Fig. 7.10 Click Install for Anaconda

After the installation of Anaconda is complete, you can go to the Windows start menu and select the Anaconda Prompt as in Fig. 7.11

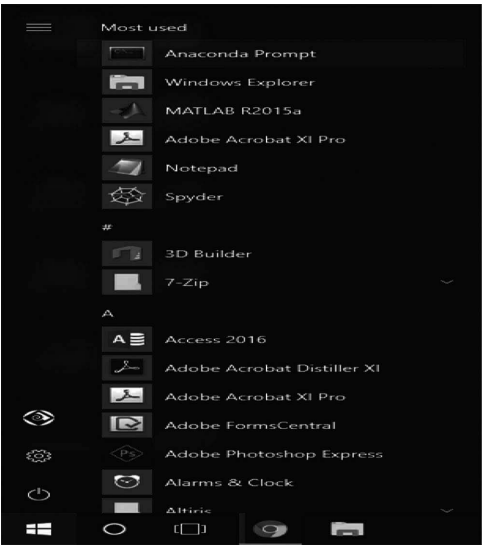


Fig. 7.11 Anaconda search in system

This opens the Anaconda Prompt. Anaconda is the Python distribution and the Anaconda Prompt is a command line shell (a program where you type in commands instead of using a mouse). The black screen and text that makes up the Anaconda Prompt doesn't look like much, but it is really helpful for problem solvers using Python.

At the Anaconda prompt, type `python` and hit [Enter]. The `python` command starts the Python interpreter, also called the Python REPL (for Read Evaluate Print Loop). you will able to access the python from terminal of anaconda prompt. From the anaconda prompt you can install all the libraries of python which are required in several projects to build.

On MacOS

Visit the Anaconda downloads page. Go to the following link: [Anaconda.com/downloads](https://anaconda.com/downloads)

Select MacOS and download the .pkg installer

In the operating systems box, select [MacOS]. Then download the most recent Python 3 distribution (at the time of this writing the most recent version is Python 3.6) graphical installer by clicking the Download link. Python 2.7 is legacy Python. For problem solvers, select the most recent Python 3 version as in Fig. 7.12.

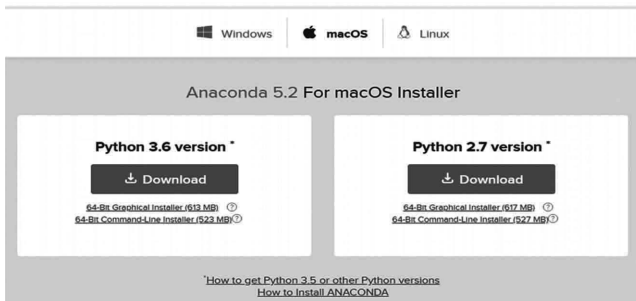


Fig. 7.12 Selecting version for Mac OS

Next Navigate to the Downloads folder and double-click the .pkg installer file you just downloaded. It may be helpful to order the contents of the Downloads folder by date to find the .pkg file.

Follow the installation instructions. It is advised that you install Anaconda for the current user and that Anaconda is added to your PATH.

Once Anaconda is installed, you need to load the changes to your PATH environment variable in the current terminal session.

Open the MacOS Terminal and type:

```
$ cd ~
$ source.bashrc
```

Open the MacOS Terminal and type:

```
$ python
```

You should see something like

```
Python 3.6.3 | Anaconda Inc. |
```

At the Python REPL (the Python >>> prompt) try:

```
>>> import this
```

If you see the Zen of Python, the installation was successful. Exit out of the Python REPL using the command `exit ()`. Make sure to include the double parenthesis `()` after the exit command.

```
>>> exit ()
```

On Linux

Go to the following link: [Anaconda.com/downloads](https://anaconda.com/downloads). On the downloads page, select the Linux operating system as in Fig. 7.13.



Fig. 7.13 Linux version

In the Python 3.6 Version* box, right-click on the [64-Bit (x86) Installer] link. Select [copy link address] as shown in Fig. 7.14



Fig. 7.14 Copy the Downloading link of anaconda

Now that the bash installer (.sh file) link is stored on the clipboard, use `wget` to download the installer script. In a terminal, `cd` into the home directory and make a new directory called `tmp`. `cd` into `tmp` and use `wget` to download the installer. Although the installer is a bash script, it is still quite large and the download will not be immediate (Note the link below includes `<release>`. the specific release depends on when you download the installer).

```
$ cd ~
$ mkdir tmp
$ cd tmp
$ https://repo.continuum.io/archive/Anaconda3<release>.sh
```

With the bash installer script downloaded, run the .sh script to install Anaconda3. Ensure you are in the directory where the installer script downloaded:

```
$ ls
Anaconda3-5.2.0-Linux-x86_64.sh
Run the installer script with bash.
$ bash Anaconda3-5.2.0-Linux-x86_64.sh
```

Accept the Licence Agreement and allow Anaconda to be added to your PATH. By adding Anaconda to your PATH, the Anaconda distribution of Python will be called when you type `$ python` in a terminal.

Now that Anaconda3 is installed and Anaconda3 is added to our PATH, source the .bashrc file to load the new PATH environment variable into the current terminal session. Note the .bashrc file is in the home directory. You can see it with `$ ls -a`.

```
$ cd ~  
$ source.bashrc
```

To verify the installation is complete, open Python from the command line:

```
$ python  
Python 3.6.5 |Anaconda, Inc.|  
[GCC 7.2.0] on Linux  
Type "help", "copyright", "credits" or "license" for more  
information.  
>>>
```

If you see Python 3.6 from Anaconda listed, your installation is complete. To exit the Python REPL, type:

```
>>> exit ()
```

Using Jupyter

The simplest way to install Jupyter notebooks is to download and install the Anaconda distribution of Python. The Anaconda distribution of Python comes with Jupyter notebook included and no further installation steps are necessary.

Below are additional methods to install Jupyter notebooks if you are not using the Anaconda distribution of Python.

Installing Jupyter on Windows using the Anaconda Prompt

To install Jupyter on Windows, open the Anaconda Prompt and type:

```
> conda install jupyter
```

Type `y` for yes when prompted. Once Jupyter is installed, type the command below into the Anaconda Prompt to open the Jupyter notebook file browser and start using Jupyter notebooks.

```
> jupyter notebook
```

Installing Jupyter on MacOS

To install Jupyter on MacOS, open the MacOS terminal and type:

```
$ conda install jupyter
```

Type `y` for yes when prompted.

If conda is not installed, the Anaconda distribution of Python can be installed, which will install conda for use in the MacOS terminal.

Problems can crop up on MacOS when using the MacOS provided system version of Python. Python packages may not install on the system version of Python properly.

Moreover, packages which do install on the system version of Python may not run correctly. It is therefore recommended that MacOS users install the Anaconda distribution of Python or use homebrew to install a separate non-system version of Python.

To install a non-system version of Python with homebrew, key the following into the MacOS terminal. See the homebrew documentation at <https://brew.sh>.

```
$ brew install Python
```

After homebrew installs a non-system version of Python, pip can be used to install Jupyter.

```
$ pip install jupyter
```

Installing Jupyter on Linux

To install Jupyter on Linux, open a terminal and type:

```
$ conda install jupyter
```

Type y for yes when prompted.

Alternatively, if the Anaconda distribution of Python is not installed, one can use pip.

```
$ pip3 install jupyter
```

Using google Colab

Google Collaboratory is a free online cloud-based Jupyter notebook environment that allows us to train our machine learning and deep learning models on CPUs, GPUs, and TPUs.

Colab gives us 12 hours of continuous execution time. After that, the whole virtual machine is cleared and we have to start again. We can run multiple CPU, GPU, and TPU instances simultaneously, but our resources are shared between these instances.

Getting Started with Google Colab:

Go to this link and sign up or login with your google account

<https://colab.research.google.com/as> seen in Fig. 7.15

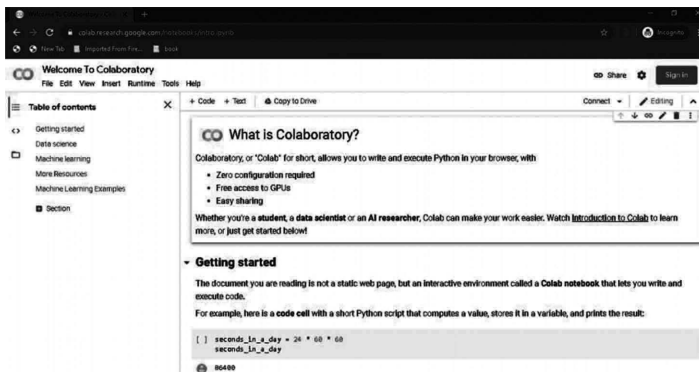


Fig. 7.15 Google Collaboratory

After login you will see this screen as in Fig. 7.16

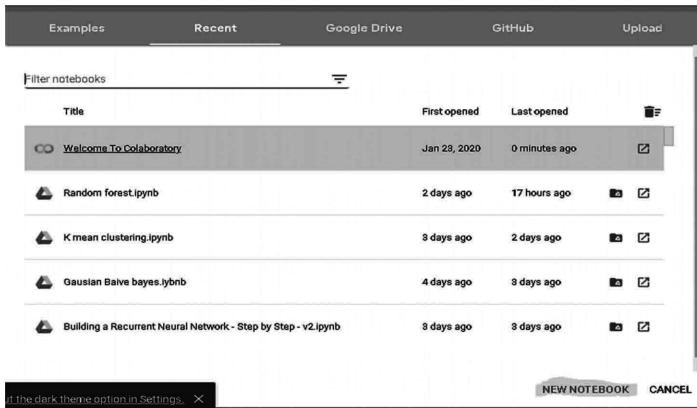


Fig. 7.16 Google Collaboratory Homepage

Click on the NEW NOTEBOOK button to create a new Collab notebook. You can also upload your local notebook to Collab by clicking the upload button as shown in Fig. 7.17:

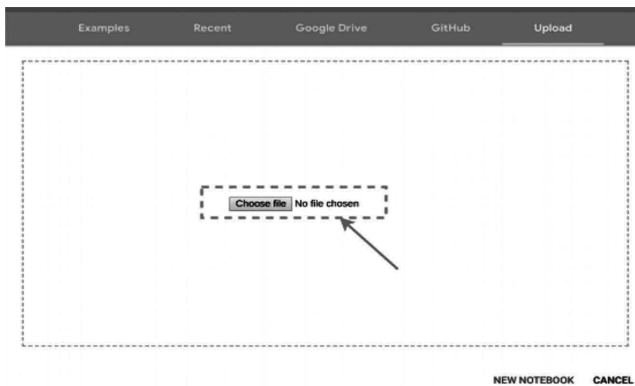


Fig. 7.17 uploading Notebook

You can also import your notebook from Google Drive or GitHub, but they require an authentication process.

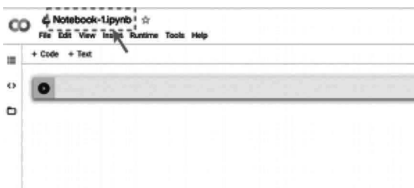


Fig. 7.18 Notebook Name change

As seen in Fig. 7.18 you can rename your notebook by clicking on the notebook name and change it to anything you want. I usually name them according to the project I'm working on.

Google Collab Runtimes – Choosing the GPU or TPU Option

The ability to choose different types of runtimes is what makes Collab so popular and powerful. Here are the steps to change the runtime of your notebook:

Click ‘Runtime’ on the top menu and select ‘Change Runtime Type’ as in Fig. 7.19

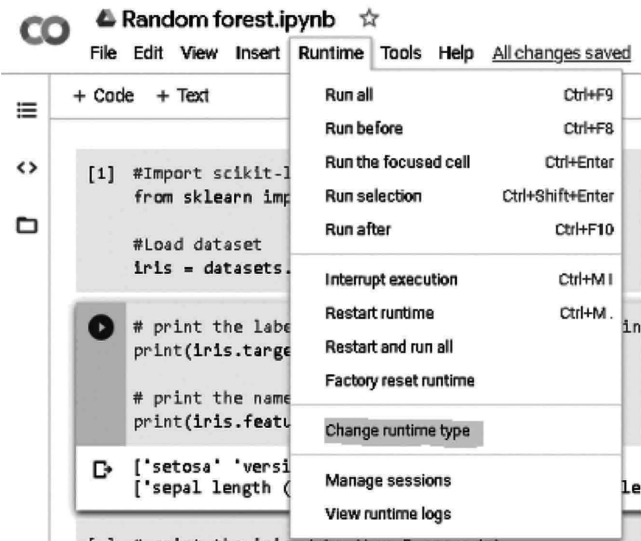
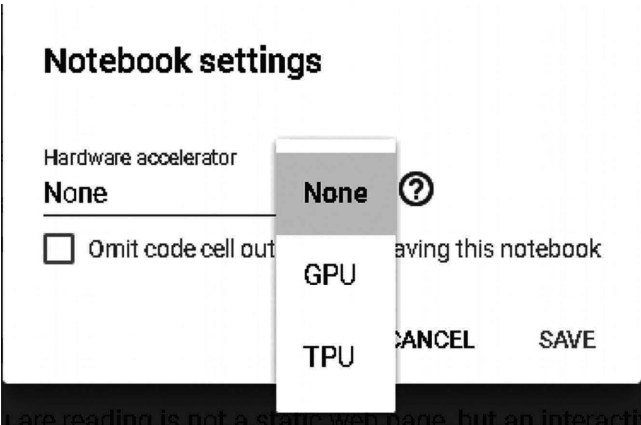


Fig. 7.19 Change Runtime Type

Here you can change the runtime according to your need as in Fig. 7.20:



g. 7.20 select Runtime type

I implore you to shut down your notebook after you have completed your work so that others can use these resources because various users share them. You can terminate your notebook like this as in Fig. 7.21:

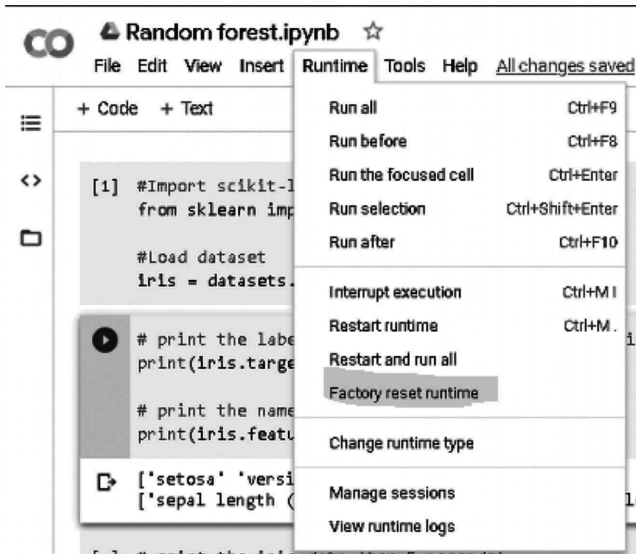


Fig. 7.21 Terminating Notebook

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Using Terminal Commands on Google Collab

You can use the Collab cell for running terminal commands. Most of the popular libraries come installed by default on Google Collab. Yes, Python libraries like Pandas, NumPy, scikit-learn are all pre-installed.

If you want to run a different Python library, you can always install it inside your Collab notebook like this:

```
!pip install library_name
```

Pretty easy, right? Everything is similar to how it works in a regular terminal. We just you have to put an exclamation (!) before writing each command like:

```
!ls
```

```
or:
```

```
!pwd
```

Uploading Files and Datasets:

Here's a must-know aspect for any data scientist. The ability to import your dataset into Collab is the first step in your data analysis journey.

The most basic approach is to upload your dataset to Collab directly as in Fig. 7.22:

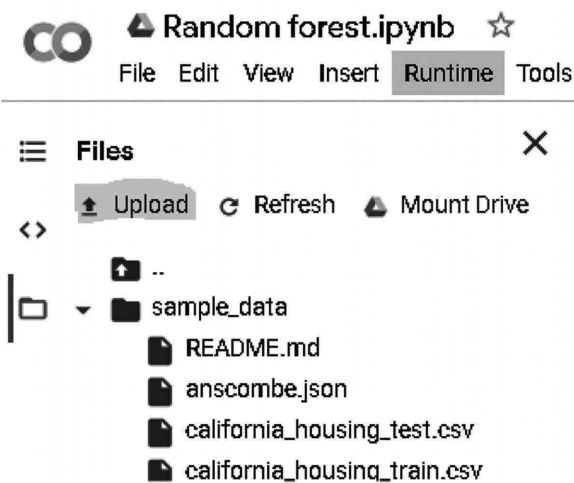


Fig. 7.22 Upload data

You can use this approach if your dataset or file is very small because the upload speed in this method is quite low. Another approach that I recommend is to upload your dataset to Google Drive and mount your drive on Collab. You can do this in just one click of your mouse as in Fig. 7.23:

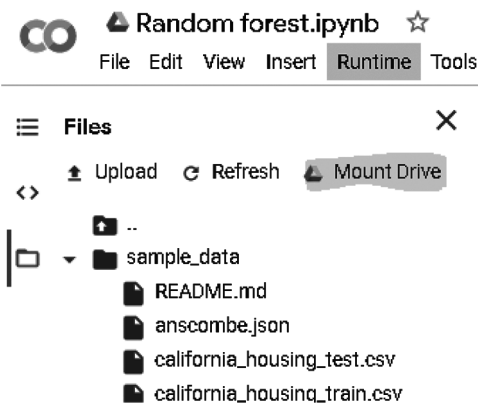


Fig. 7.23 Mounting Google drive

You can also upload your dataset to any other platform and access it using its link. I tend to go with the second approach more often than not (when feasible).

Saving Your Notebook:

All the notebooks on Colab are stored on your Google Drive. The best thing about Colab is that your notebook is automatically saved after a certain time period and you don't lose your progress.

If you want, from Fig. 7.24 you can export and save your notebook in both *.py and *.ipynb formats:

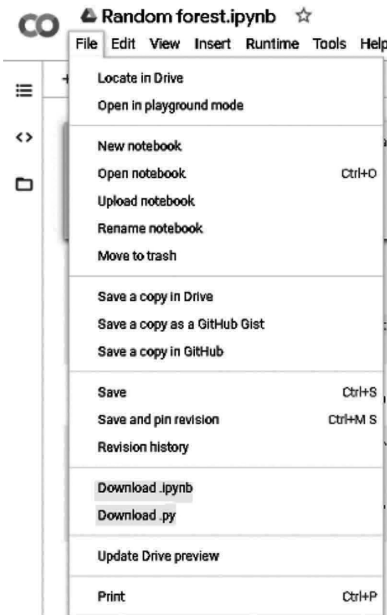


Fig. 7.24 Saving the Notebook

Not just that, you can also save a copy of your notebook directly as seen in Fig. 7.25 on GitHub, or you can create a GitHub Gist:

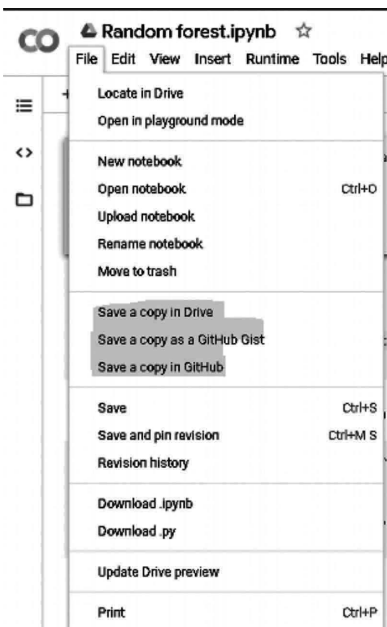


Fig. 7.25 Save Notebook

Sharing Your Notebook:

Google Collab also gives us an easy way of sharing our work with others. This is one of the best things about Collab as seen in Fig. 7.26:



Fig. 7.26 Sharing Notebook

Just click the Share button, and it gives us the option of creating a shareable link that we can share through any platform. As in Fig. 7.27 you can also invite others using their email IDs. It's exactly the same as sharing a Google Doc or Google Sheet. The intricacies and simplicity of Google's ecosystem are astounding!



Fig. 7.27: Link generating

Summary

Tools for building a project in Artificial Intelligence as discussed can be like python, jupyter, google Collaboratory. By studying all these tools we can clearly say that now we can easily work on these environment for biding a real time project which can be useful. Rather than all these you can go foe amazon EC2 server which will give you 1 year free trail to host any model or project.